INVESTIGATION OF THE ACTUATOR DISK MODEL FOR SIMULATION OF SHORT INTAKE AERODYNAMICS IN CROSSWIND

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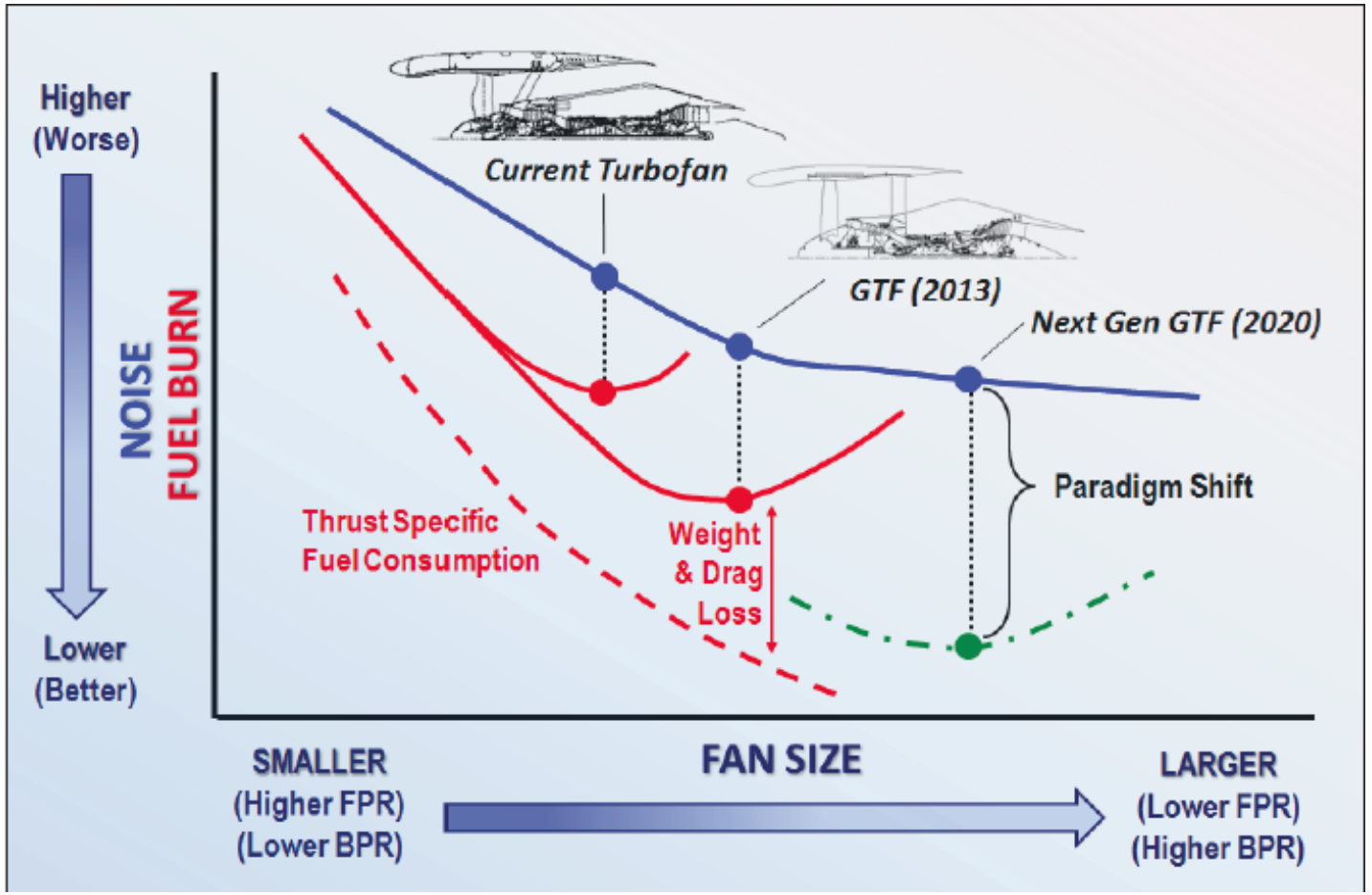
17th Overset Grid Symposium 2026
State College, PA, Sept 2026

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NEAR-TERM ENGINE IMPROVEMENTS WILL FEATURE HIGHER BYPASS RATIOS & LOWER FAN PRESSURE RATIOS

- Higher bypass ratios (BPR) and lower fan pressure ratios (FPR) enable thrust to be generated more efficiently
- Increasing BPR increases nacelle weight and drag, and engine core weight
- Aggressive nacelle designs needed to mitigate barriers to BPR increases



Hughes, Chris. "The promise and challenges of ultra high bypass ratio engine technology and integration." *49th AIAA Aero Sciences Meeting*. January 2011.

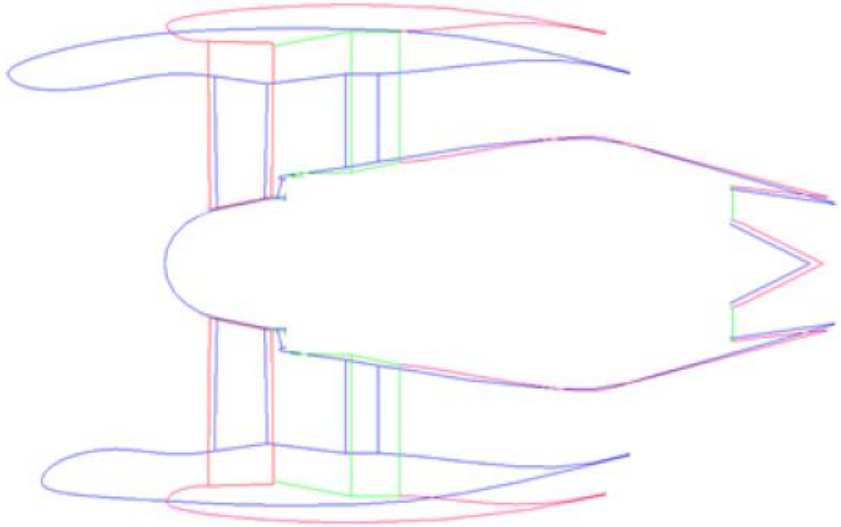
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SHORT INLETS OFFER A SOLUTION WITH SOME CONSIDERATIONS

- Reducing intake (L/D < 0.5) and exhaust lengths can reduce nacelle weight and drag

- Short intakes reduce diffusion capabilities, increases fan exposure to flow distortions (e.g. sideslip separation, ground vortex ingestion, etc.)

- Distortions can have adverse effects on engine performance, operability, and safety



Hughes, Chris. "The promise and challenges of ultra high bypass ratio engine technology and integration." *49th AIAA Aero Sciences Meeting*. January 2011.



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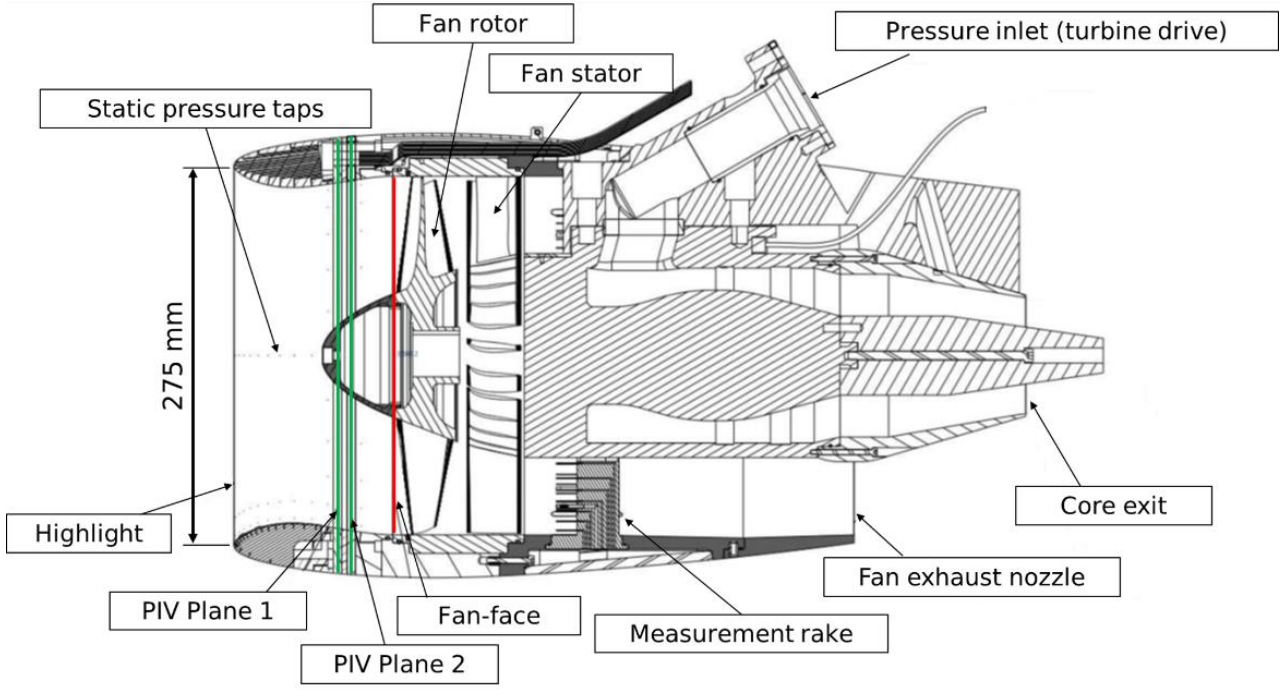
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AIAA PROPULSION AERODYNAMICS WORKSHOP 07

NASA

- Scitech 2026: assess CFD capabilities for inlet aerodynamics in crosswind w/ propulsor using NIFTI geometry
  - *Non-Intrusive Flow distortion measurements within a Turbofan Intake*
  - Features short intake (L/D = 0.452)
  - Designed to fit into CRM nacelle
- Open geometry w/ experimental data* (PIV image planes)

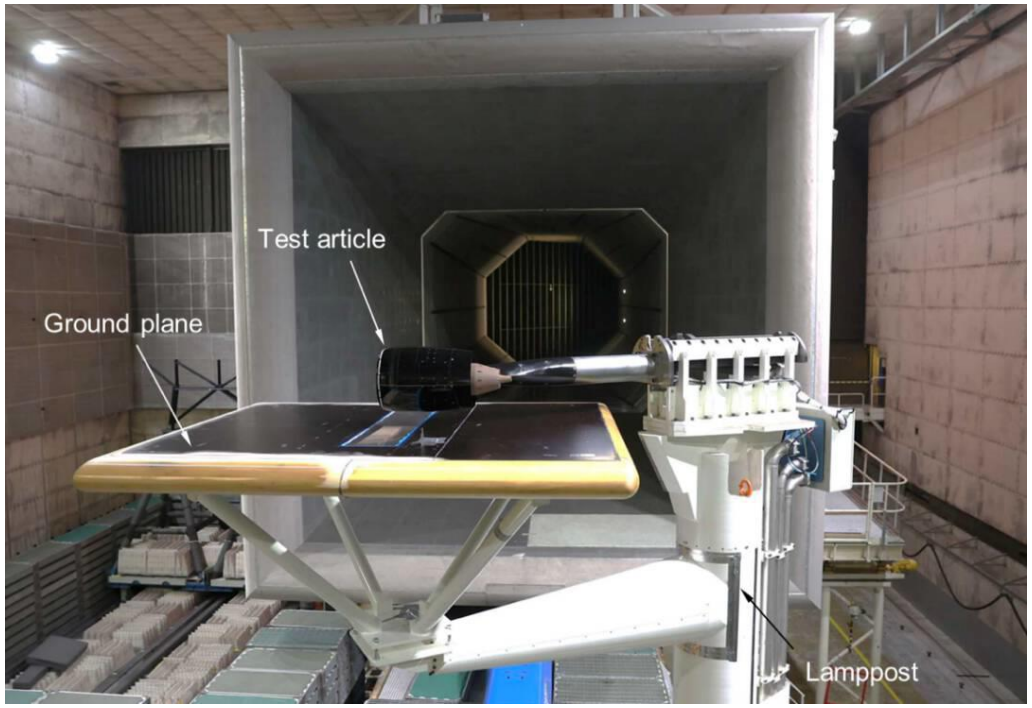
Component	Blade/Vane Count
Fan Rotor	19
Stator	25

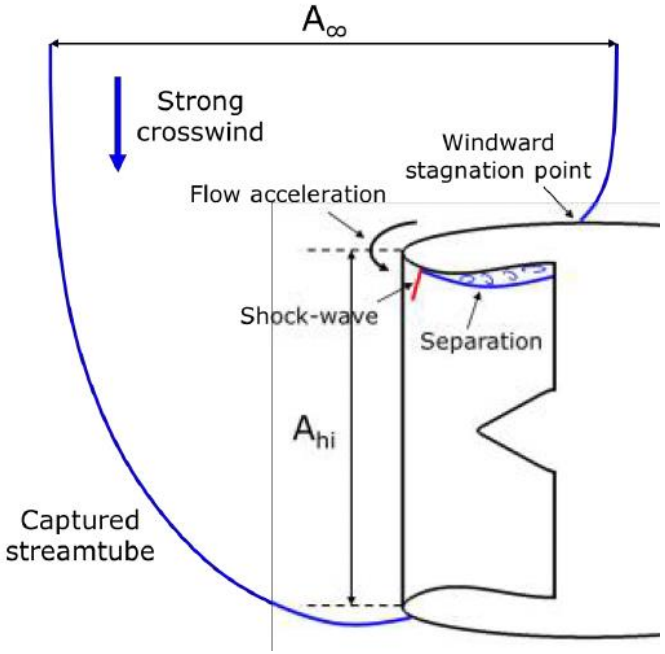


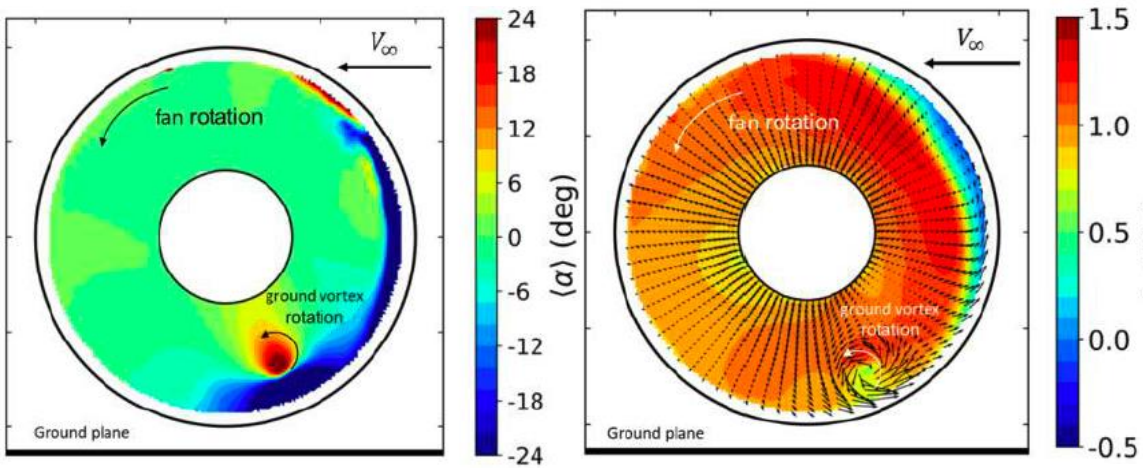
*Piovesan et al., AIAA 2025-2111. January 2025.

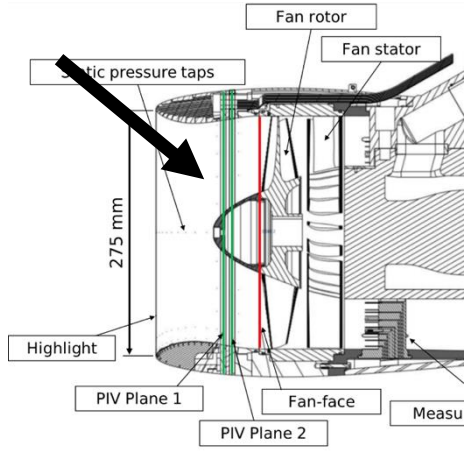
NIFTI INTAKE CROSSWIND TEST

- Conducted for CFD validation
- Subject to crosswind (0-30 kts) and ground plane







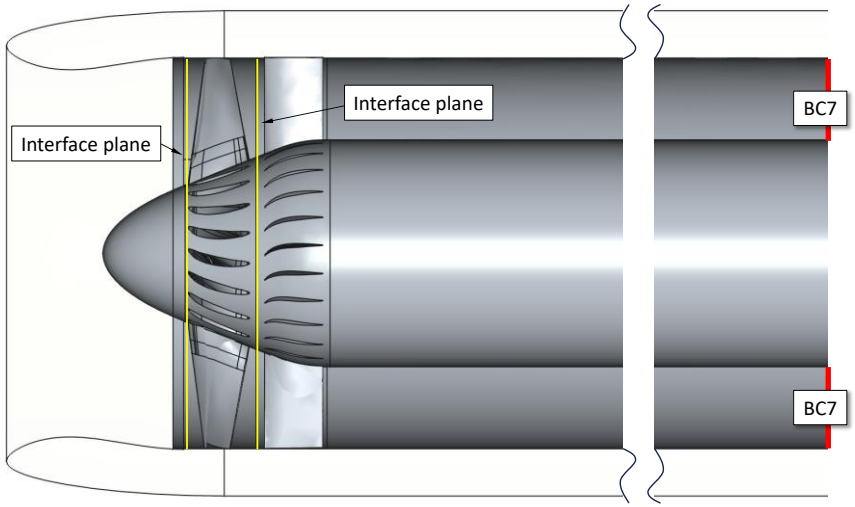


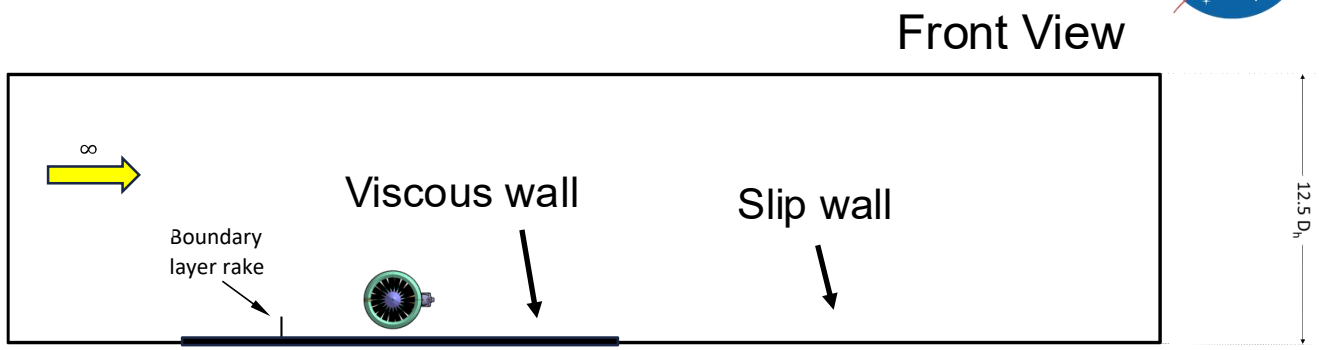
Piovesan et al., AIAA 2025-2111. January 2025.

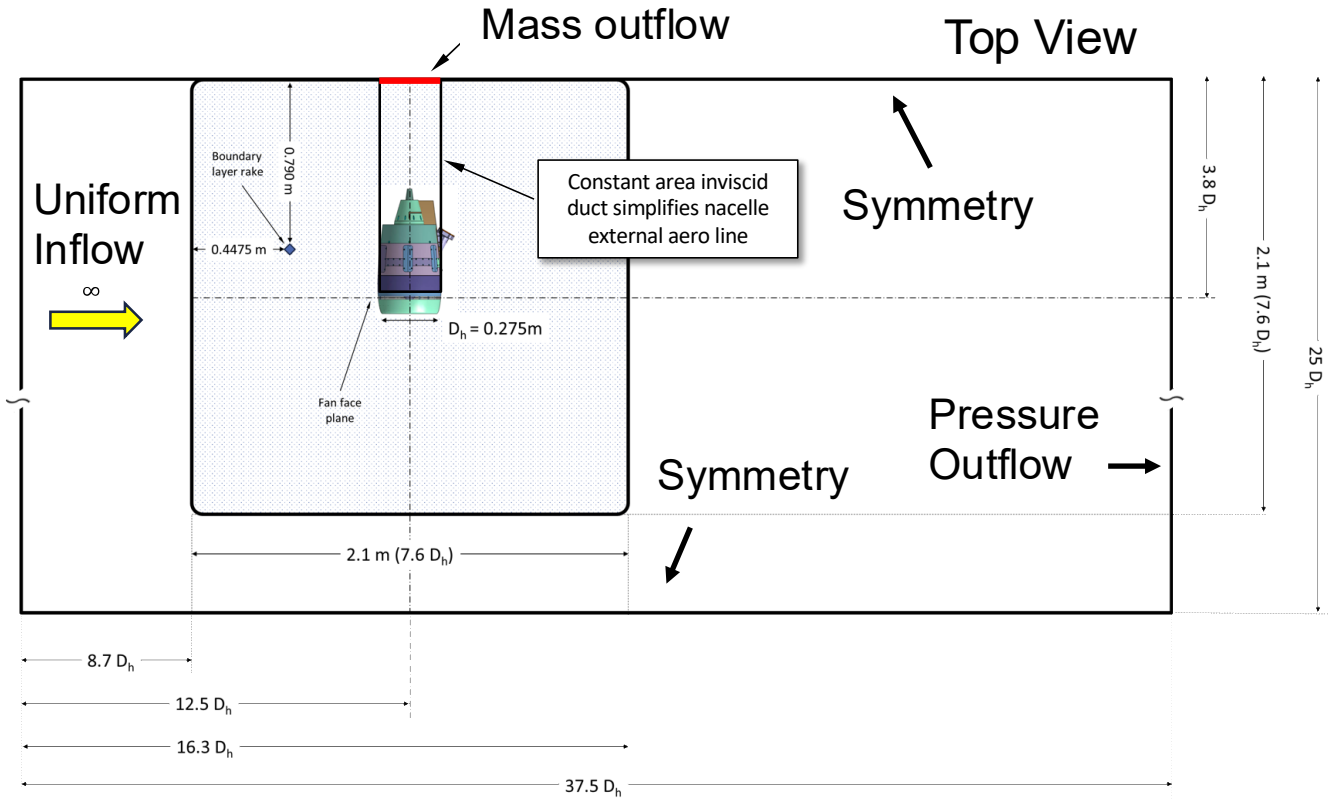
COMPUTATIONAL SETUP

- Only propulsor & intake considered inside simplified wind-tunnel environment

- Bottom surface split into viscous wall (ground-plane) and slip wall (free-air)







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WHAT DOES IT TAKE TO SUFFICIENTLY SIMULATE INLET AERODYNAMIC DISTORTIONS?

- Low-order models can be useful for where blade-resolved computations are cost-prohibitive
  - Propulsion-airframe integration analysis, installation effects, etc.

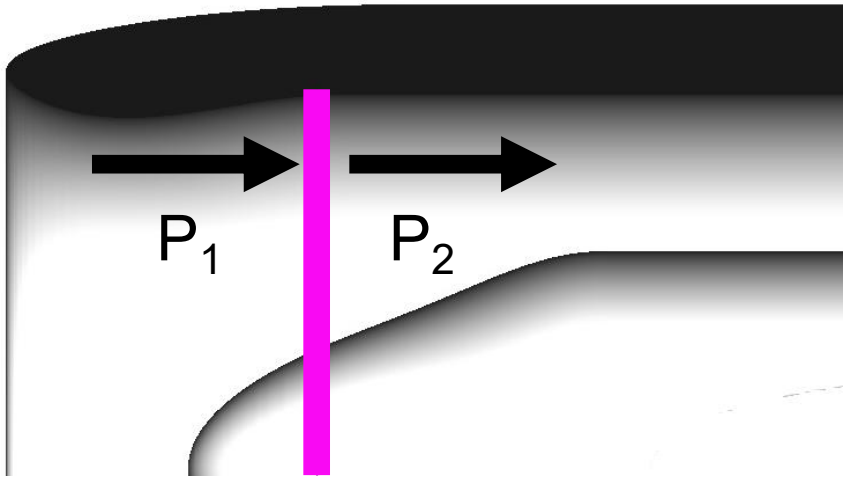
- Objective: Investigate capabilities & methodologies of low-order models for simulating intake distortion with propulsor
  - Pressure disk
  - **Rotor disk (blade element theory)**
  - Body-Force models (future work)

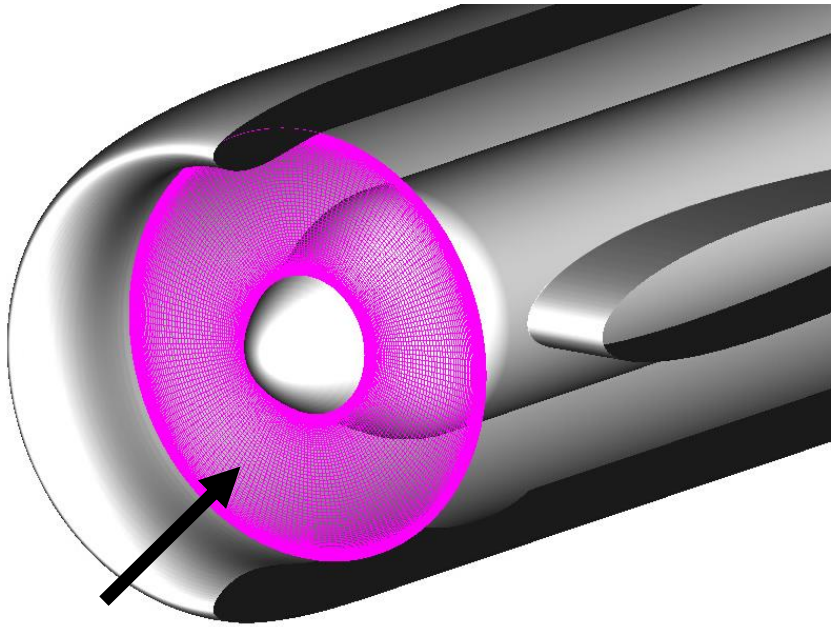
- Compare with experimental data & URANS

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PRESSURE DISK MODEL

- Simple static pressure jump across fan region $\Delta P = (P_2 - P_1)$
- Obtained from fan pressure ratios $FPR = {}^{P_{t,2}}/_{P_{t,1}}$ and isentropic flow relations
- EGVs not included (no swirl)



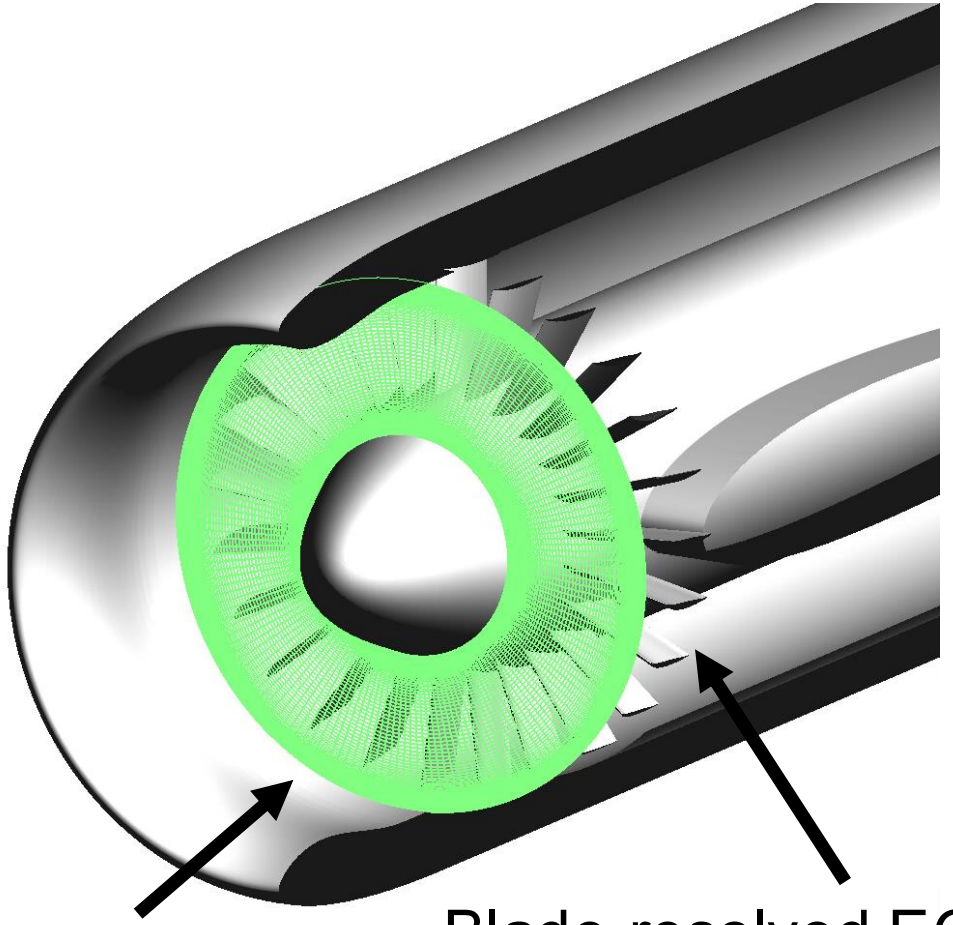


Fan pressure-disk

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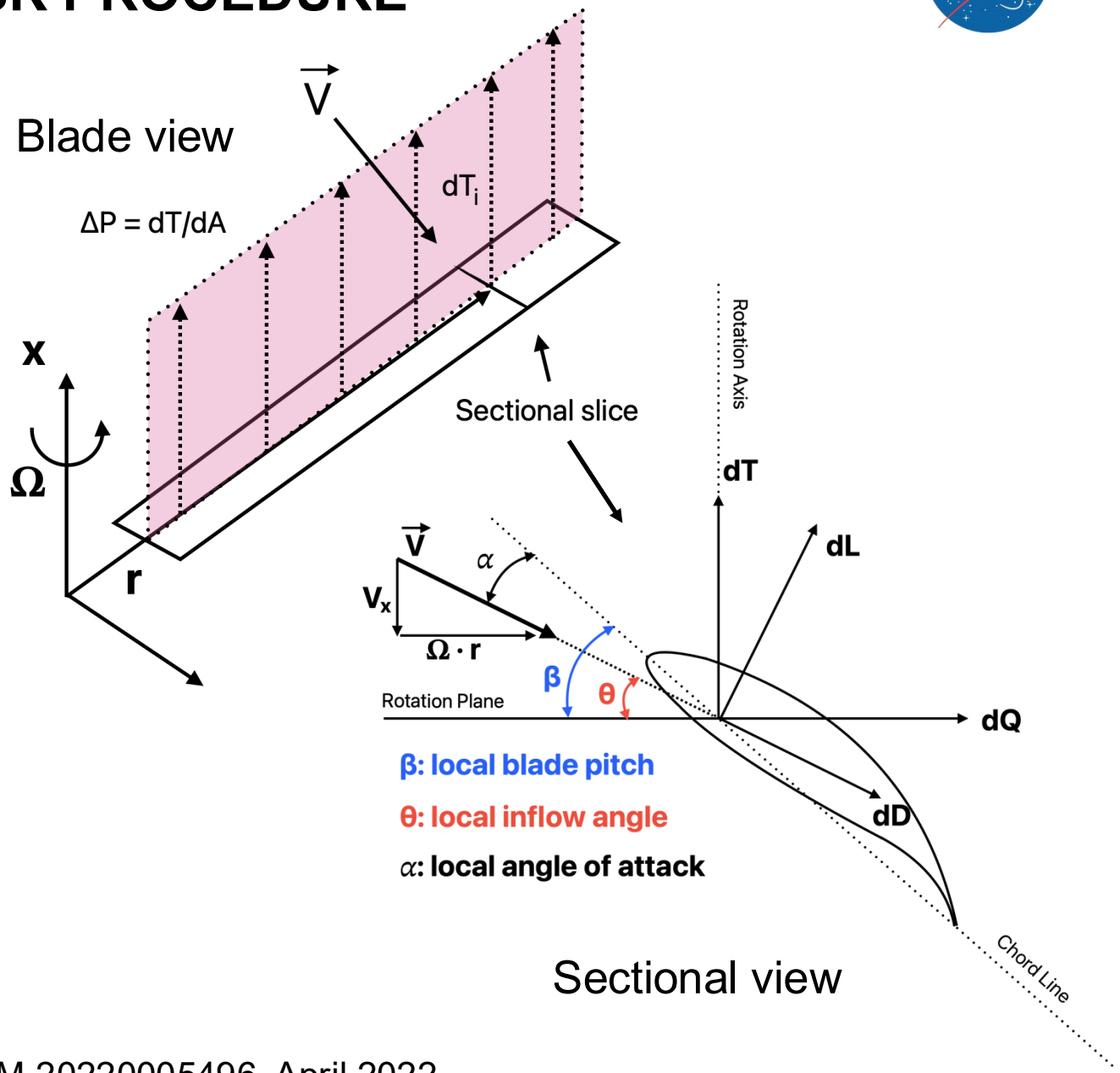
ROTOR DISK MODEL

- Applies loads from Blade Element Theory (BET) averaged across disk
- Introduces thrust and swirl effects into flow
- Can be applied to both fan rotor and exit guide-vanes (EGV) (if no azimuthal varations)



ROTOR DISK PROCEDURE

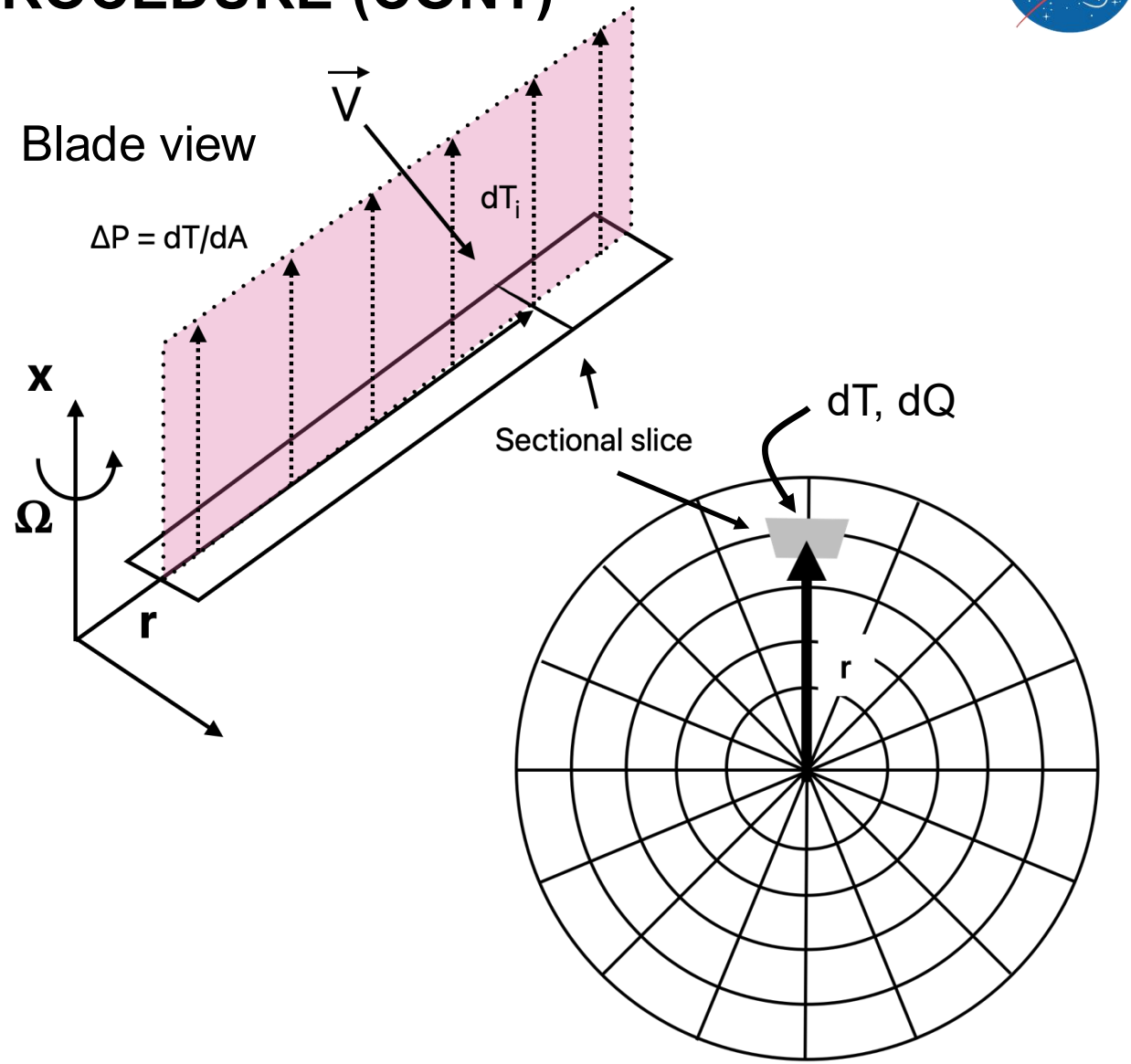
- Blade-Element Theory: Blade loading constructed from spanwise/radial sectional ”elements” (airfoil slices)
- Sectional lift and drag obtained from local inflow and airfoil tables (C81)



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ROTOR DISK PROCEDURE (CONT)

- Sectional lift and drag converted into local thrust (dT) and torque (dQ), averaged in time across rotor “disk”
- Disk thrust & torque adds source terms to momentum & energy eqns. that induce pressure rise and swirl

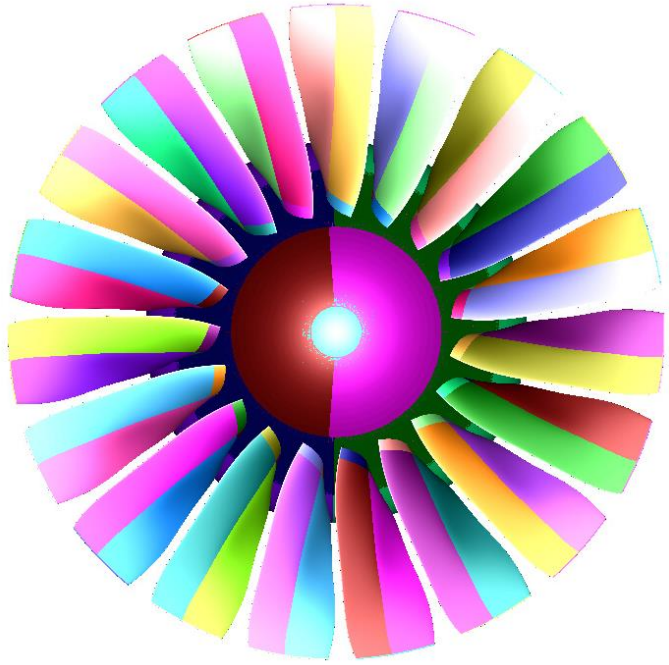


Actuator disk plane BC

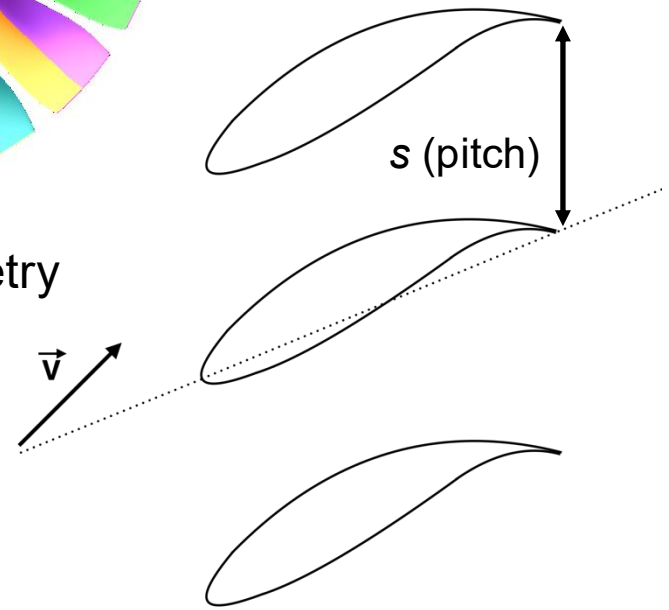
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CONSIDERATIONS FOR TURBOFANS

- Turbofan rotors have **high solidity**—blades cannot be considered in isolation
- Spinner et al (2020)*: rotor disk gives reasonable results if accounting for **cascade effects** and **pressure evolution**



NIFTI Fan Geometry



Cascade of staggered airfoils

*Spinner, S., Keller, D., Schnell, R., and Trost, M., AIAA 2020-2749. *AIAA AVIATION 2020 FORUM*.

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INVESTIGATING CASCADE EFFECTS ON ROTOR DISK MODELING (WIP)

- Low-order methods
  - Thin airfoil theory (with* & without cascade correction)
- 2D methods
  - XFOIL
  - MISES (airfoil in blade-passage, turbomachinery environment)
  - RANS CFD of airfoil (free air/blade-passage using OVERFLOW)
- 3D methods (Stich et al. 2022**)
  - Coupled fan/stator blade passage with mixing planes (LAVA, forthcoming)
  - URANS/Blade-resolved CFD (Overflow)

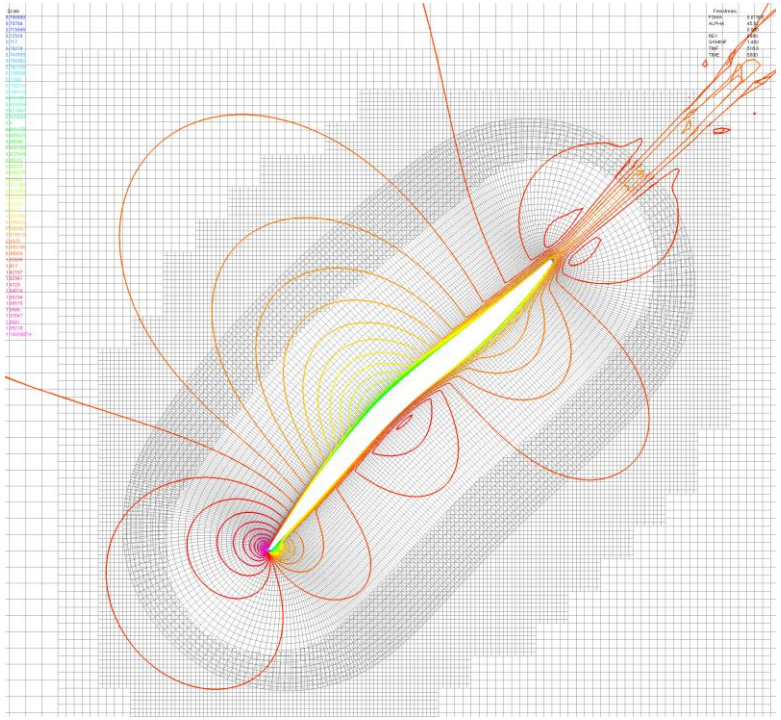
*Baddoo, P.J. and Ayton, L.J., 2018. Potential flow through a cascade of aerofoils: direct and inverse problems. *Proceedings. Mathematical, Physical, and Engineering Sciences*, *474*(2217), p.20180065.

**Stich, G. D., Fernandes, L. S., Duensing, J. C., Housman, J. A., Kenway, G. K., & Kiris, C. C. , ICCFD11-2022-0903. *ICCFD11*.

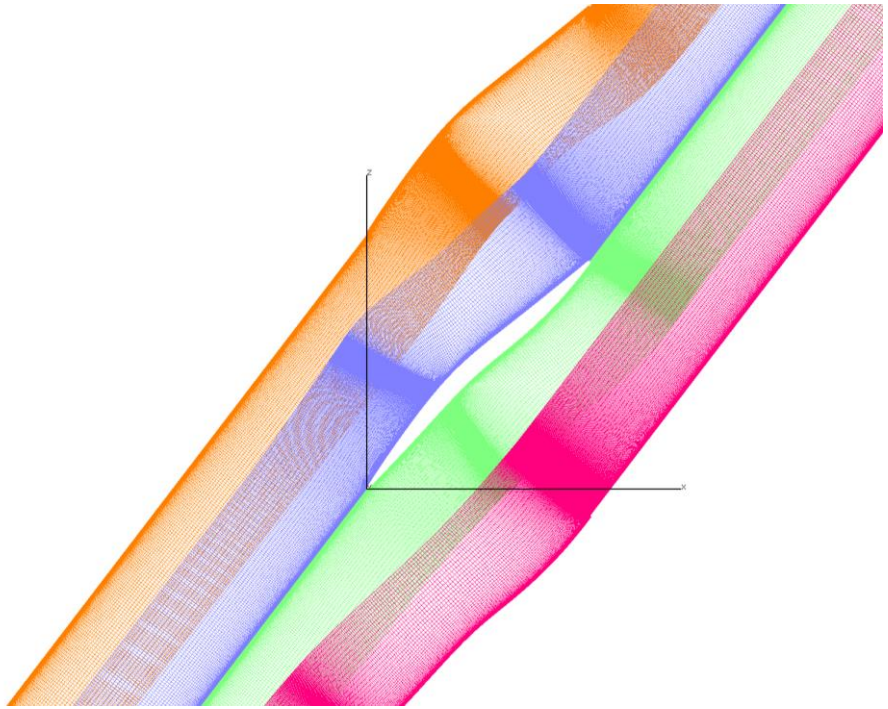
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CURRENT PROGRESS ON SECTIONAL LOADS

- Generating airfoil tables using using Overflow (WIP)



Free-air computation



Blade-passage grids

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COMPUTATIONAL EXPERIMENTS

Engine RPMs analyzed in PAW 07

Case	V_∞ (kts) Crosswind	RPM (kRPM)	$\dot{m}$ (kg/s)	FPR
2 (High RPM)	12.5	26.9	7.99	1.26
3 (Low RPM)	12.5	13.8	4.07	1.06

- Pressure disk, sans fan and stator
- Fan Rotor disk with EGV geometry included
  - Default sectional loading from OVERFLOW: symmetric thin-airfoil theory ($C_l = 2\pi\alpha$, $C_d = 0$)
- URANS w/ blade-resolved fan & stator geometries

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OVERSET MESH GENERATION

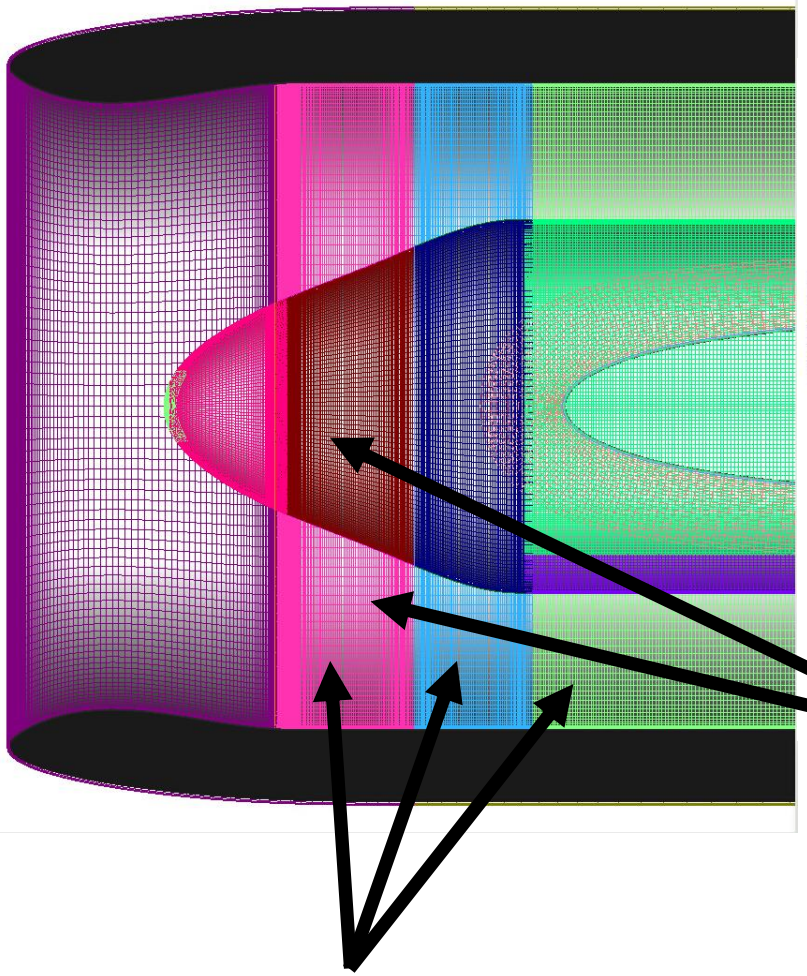
- Built via Chimera Grid Tools TCL Script Library
- Representative of a “coarse” level grid

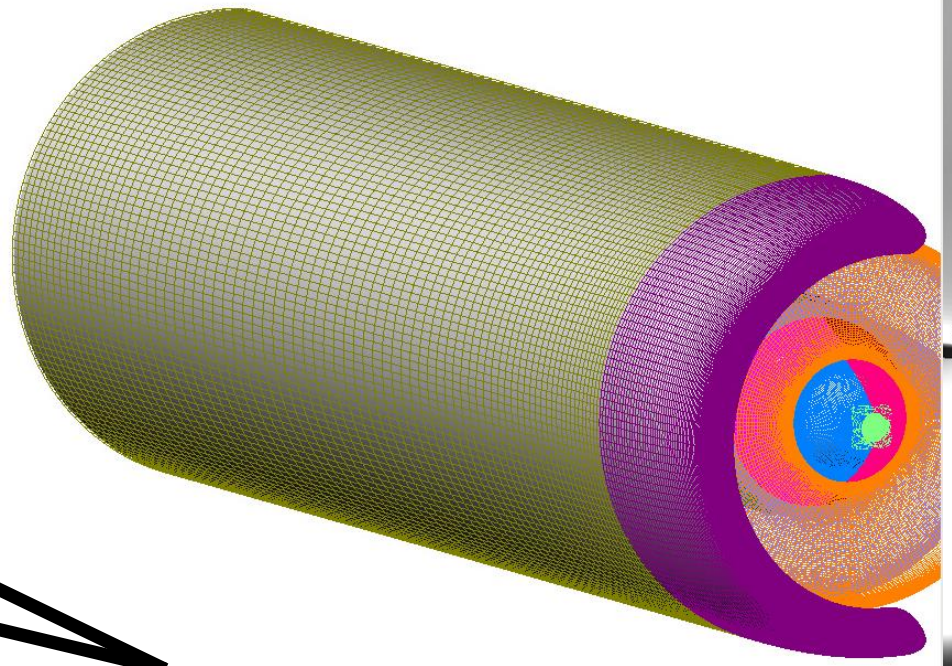
Component	P-Disk	Actuator Disk	Blade-Resolved
Core + Annular Vol.	30.9M	43.7M	56.4M
Lip + Outer Shroud	5.43M	5.43M	5.10M
Pylon	2.0M	2.0M	2.0M
Fan Blades	-	-	37.2M (1.96M/blade)
EGVs	-	45.1M (1.80M/blade)	45.1M (1.80M/blade)
Wind Tunnel	25.3M	25.3M	25.3M
Total Num. Pts.	63.7M	122M	171M
Total Grid Ct.	23	173	327

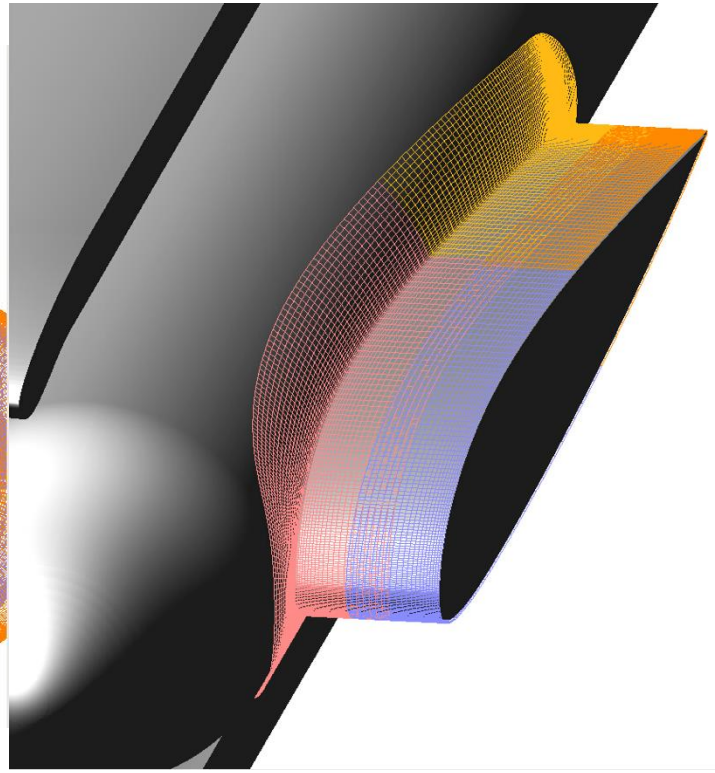
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OVERSET MESHES – NACELLE CORE, SHROUD, & PYLON

Minimum 720 pts azimuthally





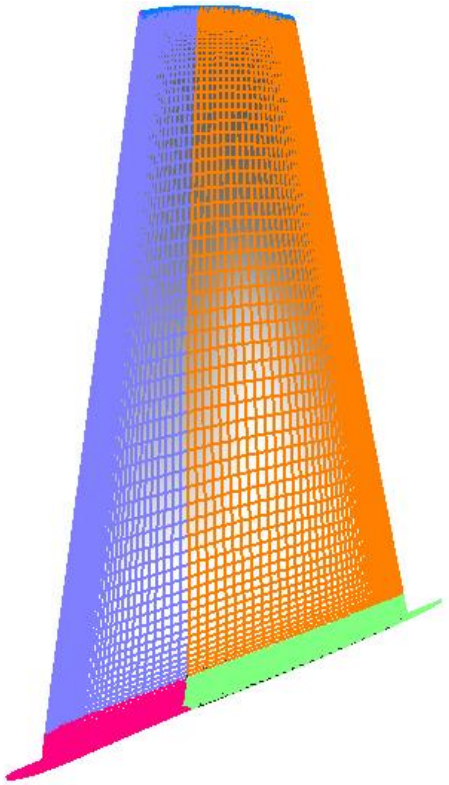


Annular volume grids

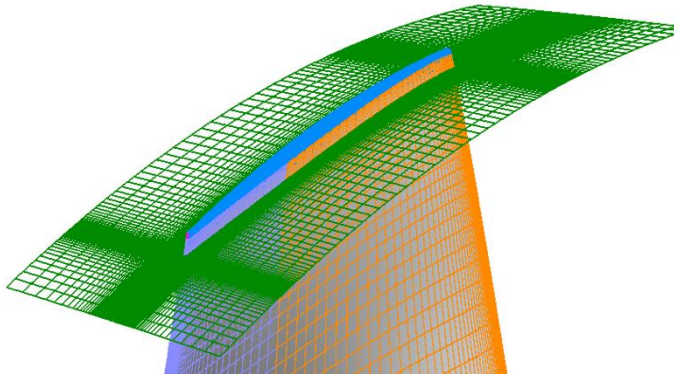
Disk models applied to this annular region

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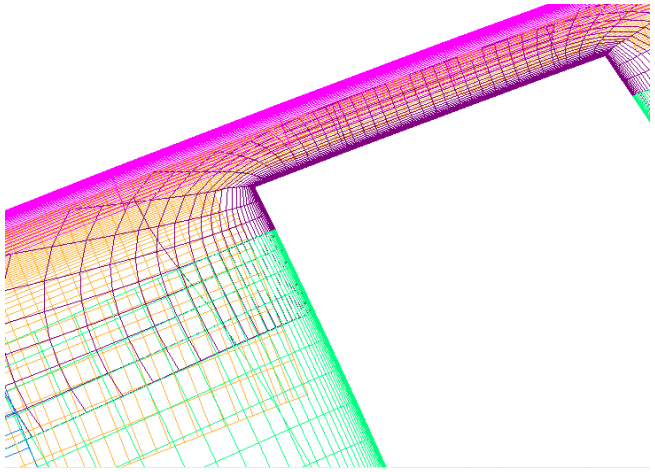
OVERSET MESHES – ROTOR AND STATOR BLADES



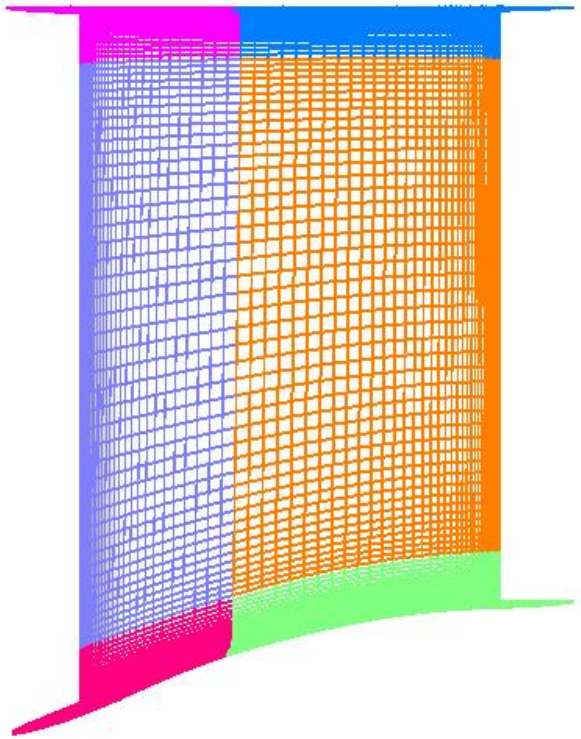
Rotor Blade Mesh
- 228 Chord-wise pts
- 169 Span-wise pts
- 42 Wall-normal pts



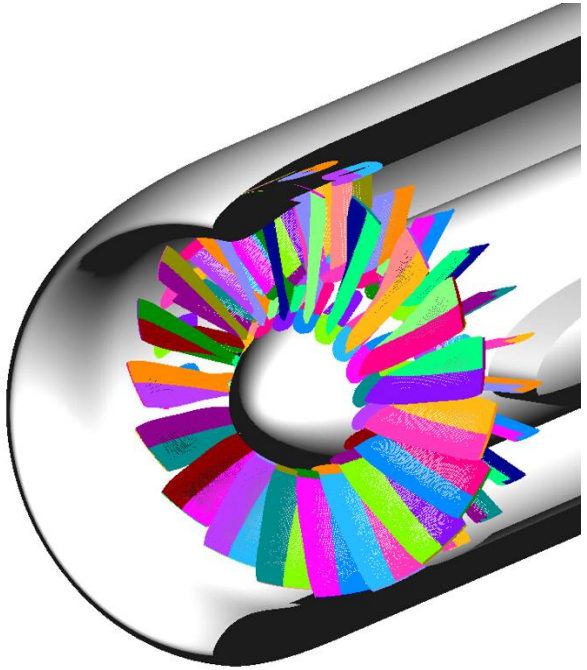
Rotor tip topology with tip box grid (green)



Overlap at blade tip/nacelle inner wall



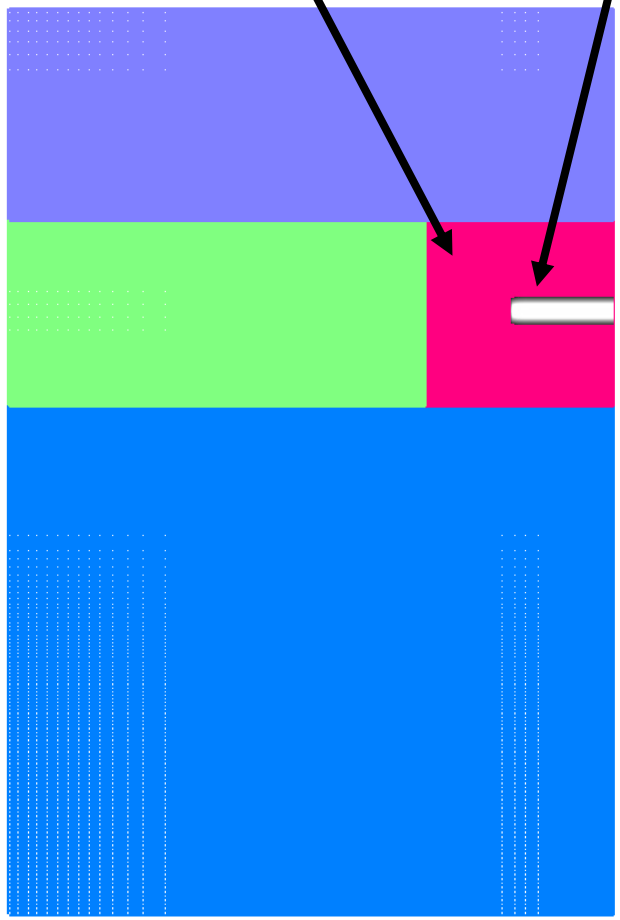
Stator Blade Mesh
- 169 Chord-wise pts
- 106 Span-wise pts
- 61 Wall-normal pts



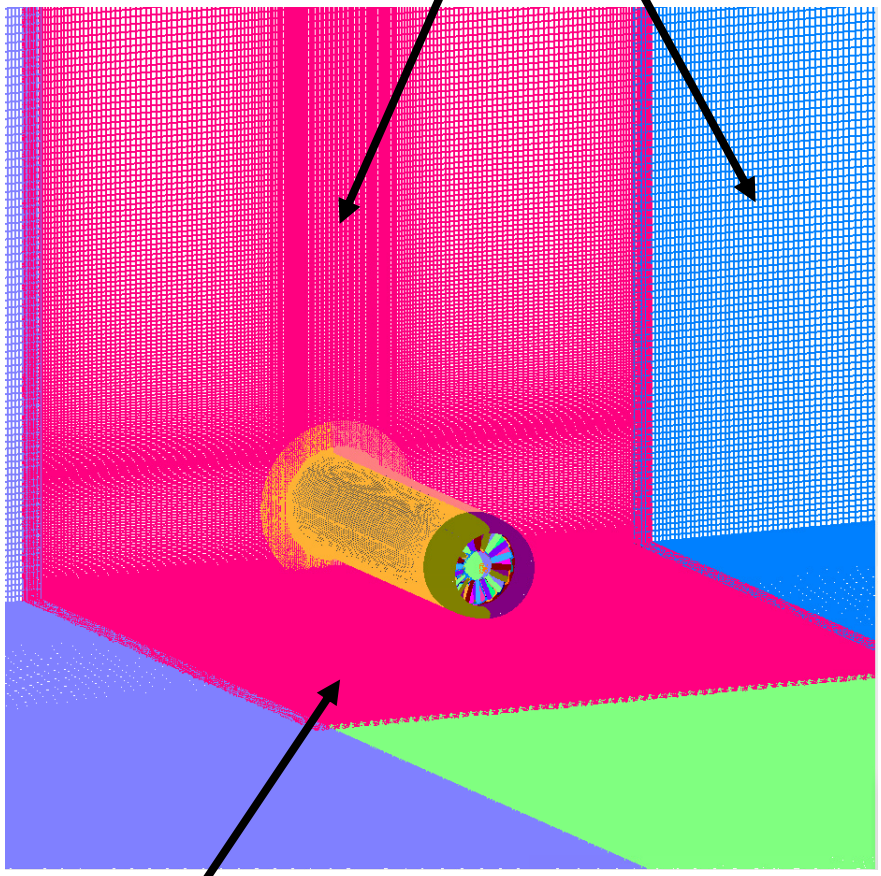
OVERSET MESHES – WIND TUNNEL

Viscous wall region (ground plane)

NIFTI Turbofan



Wind tunnel walls



Viscous wall region (ground plane)

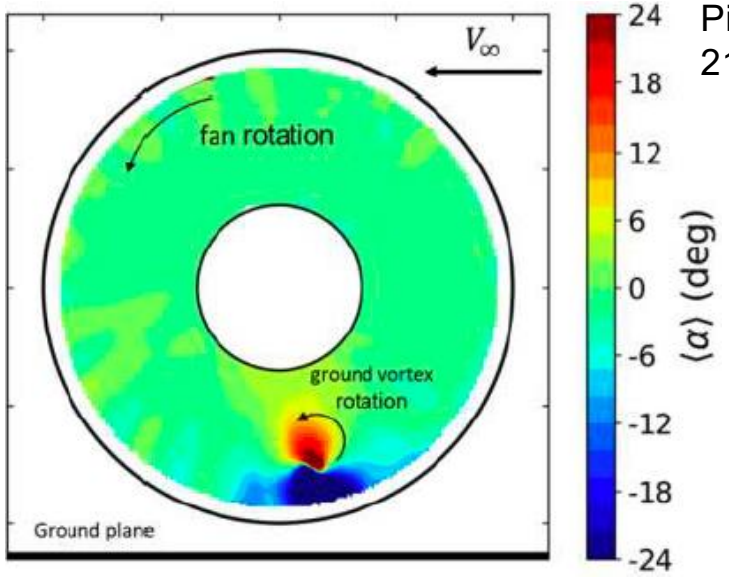
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OVERFLOW COMPUTATIONAL PROCEDURE

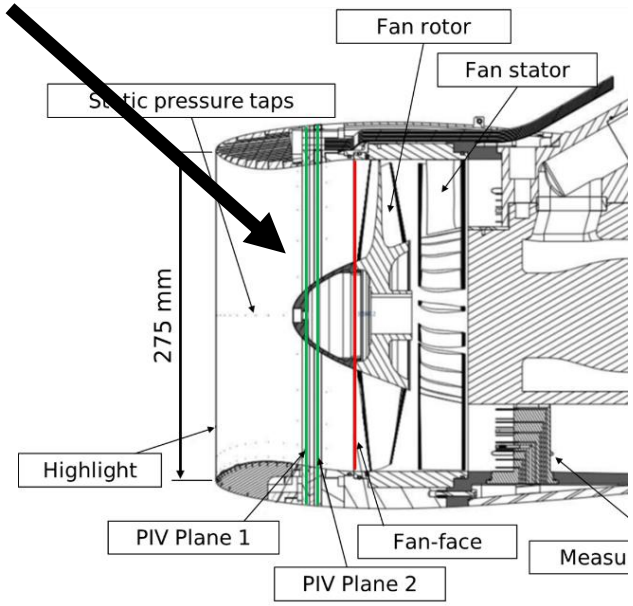
- Solver options:
  - Improved SSOR w/ Global Linear Solve
  - 3rd Order HLLE++ w/ Koren Limiter
  - Spallart-Allmaras with Rotational Correction
- Steady run procedure:
  - FMG, with CFL-ramping where applicable
  - Converge until residual drops 5 orders or QoI statistically converged (WIP)
- Unsteady run procedure:
  - Initial steady-state run with frozen blades, mass outflow BC enforced
  - Coarse time-stepping: 20+ revs (2 deg/step)
  - Fine time-stepping: ramping down over multiple revs to 0.25 deg/step (WIP)

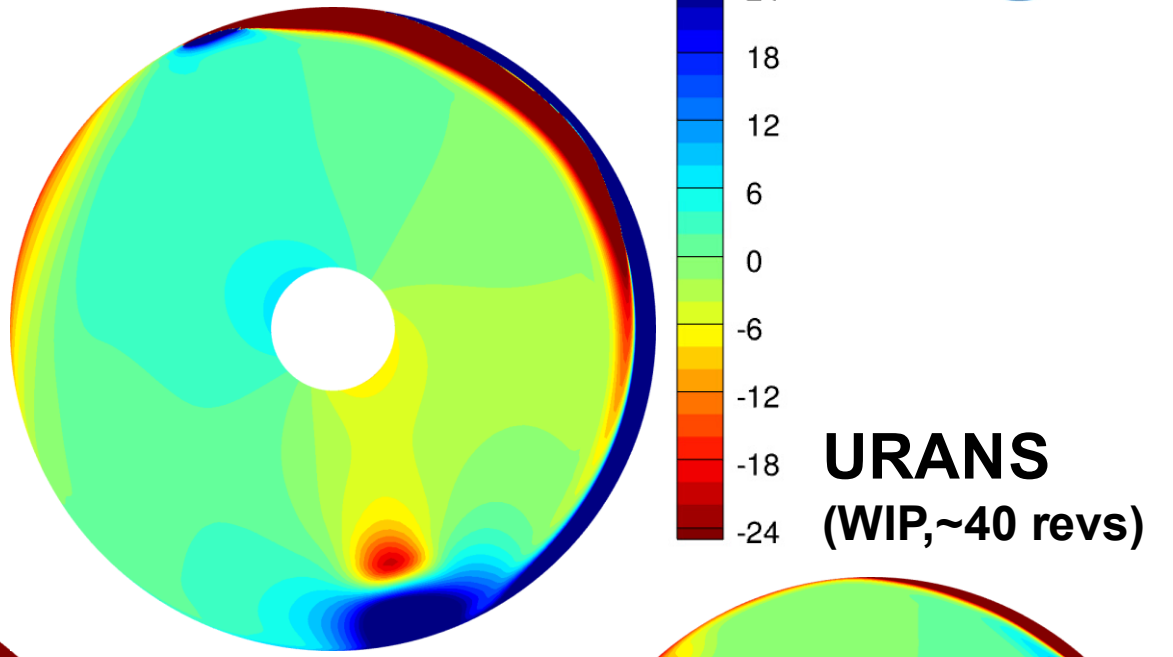
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HIGH RPM SWIRL ANGLE



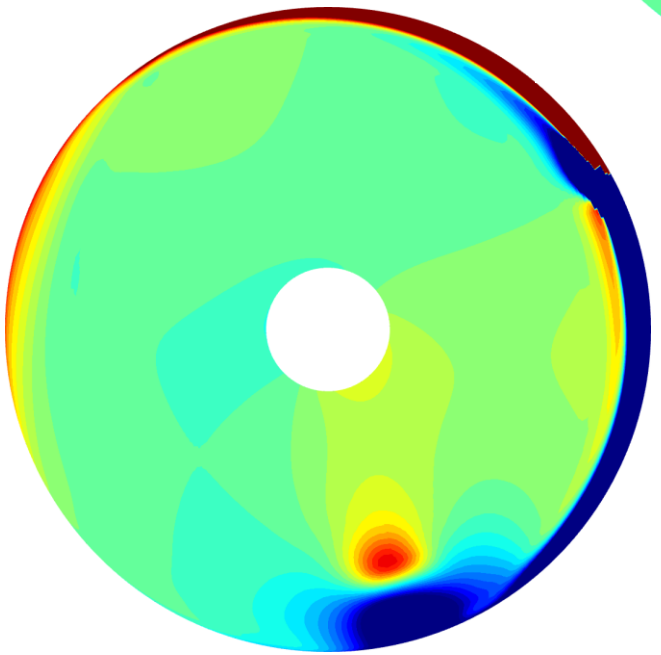
Piovesan et al., AIAA 2025-2111. January 2025.



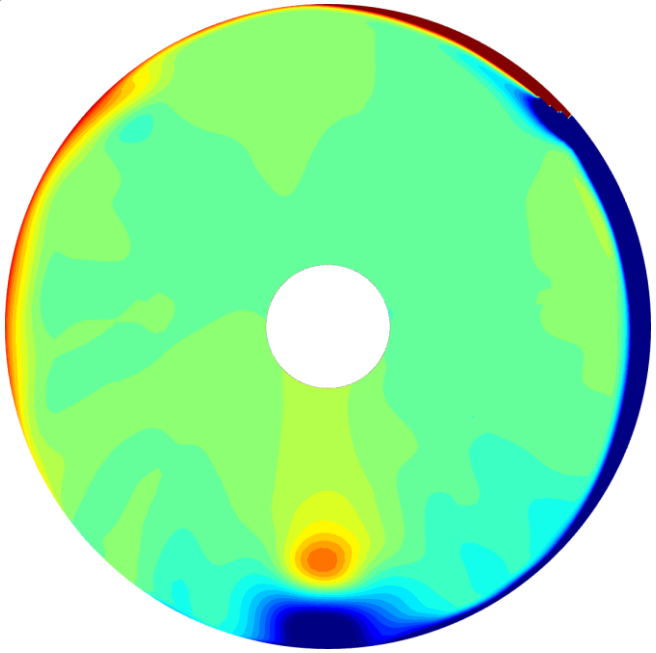


URANS
(WIP,~40 revs)

PDISK

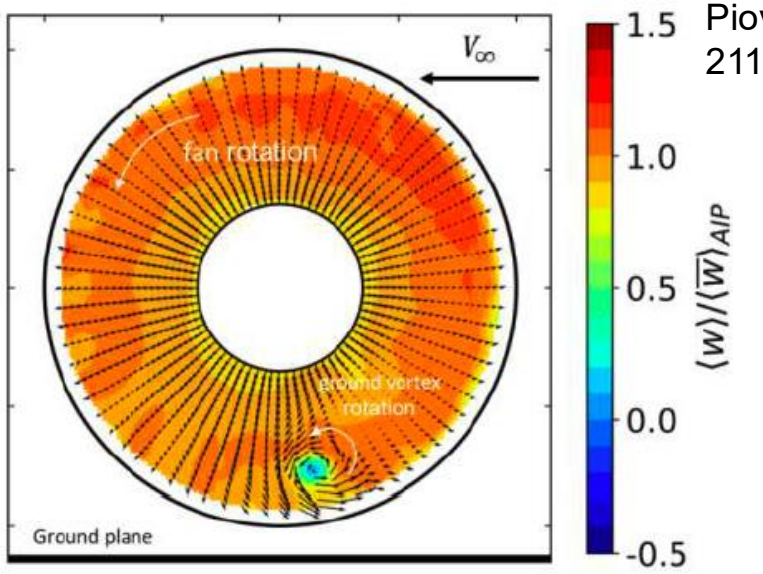


RDISK

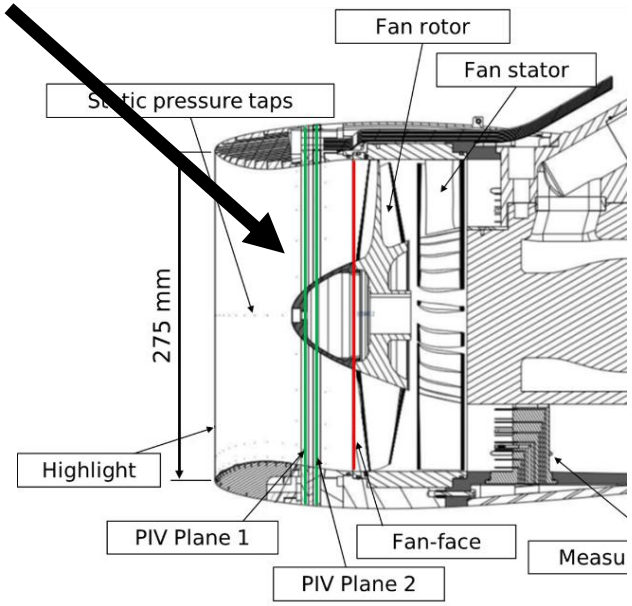


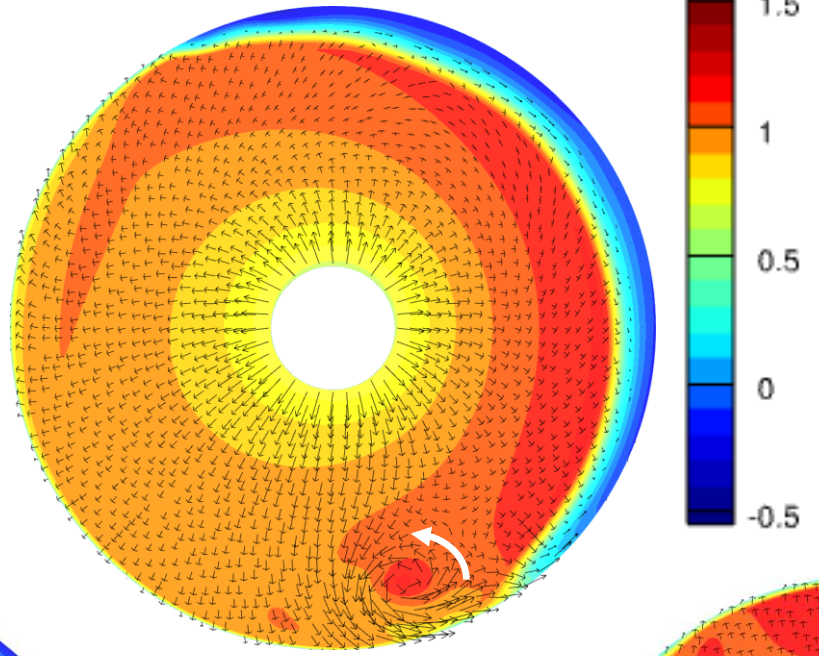
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HIGH RPM AXIAL VELOCITY

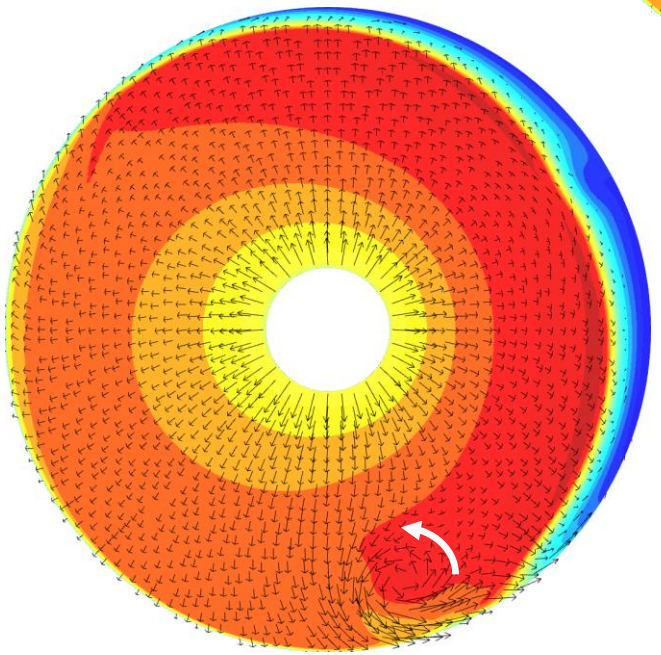


Piovesan et al., AIAA 2025-2111. January 2025.



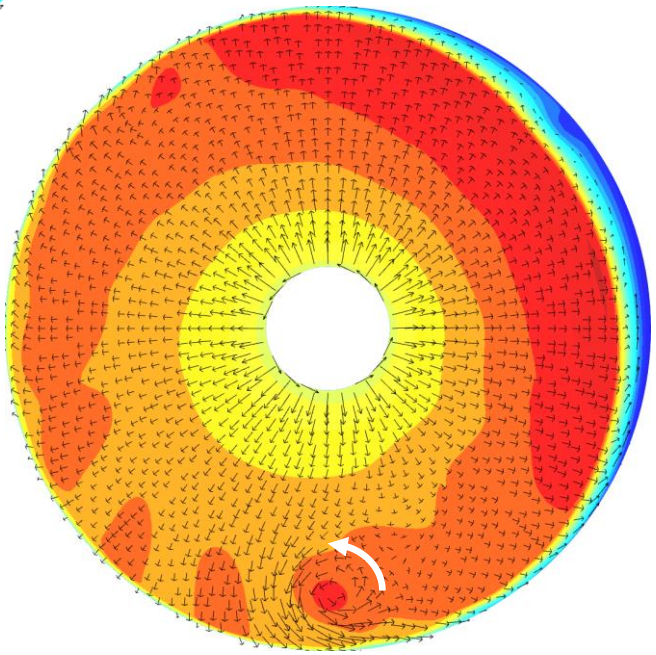


PDISK



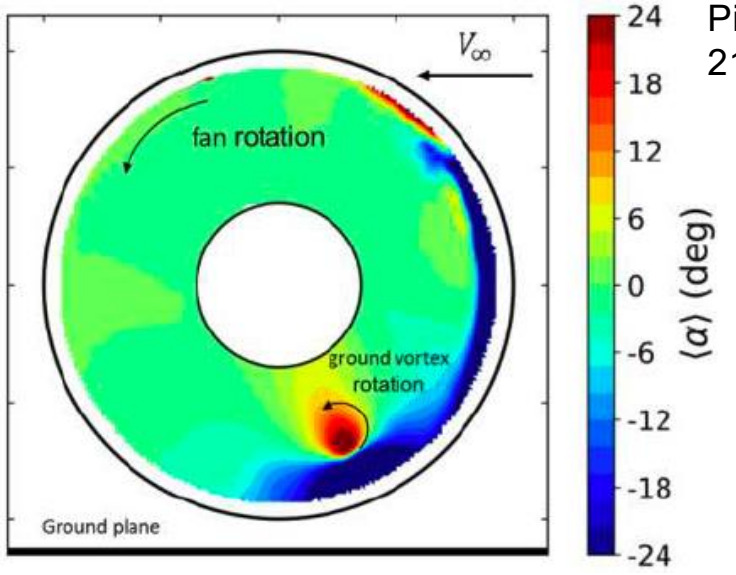
RDISK

URANS (WIP)

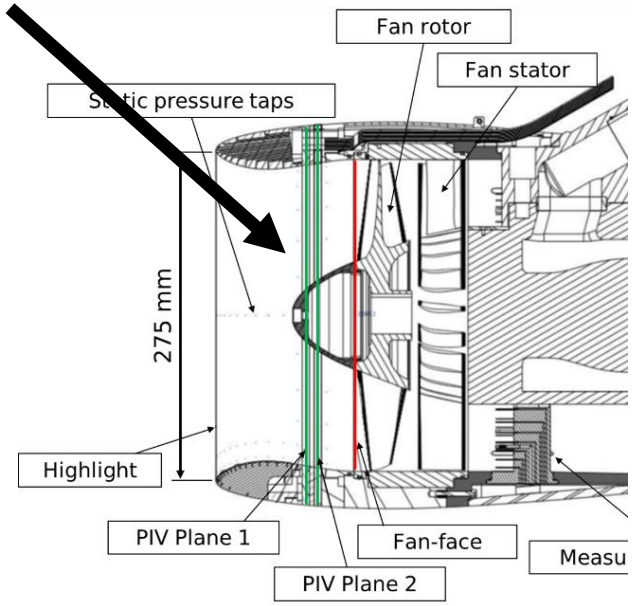


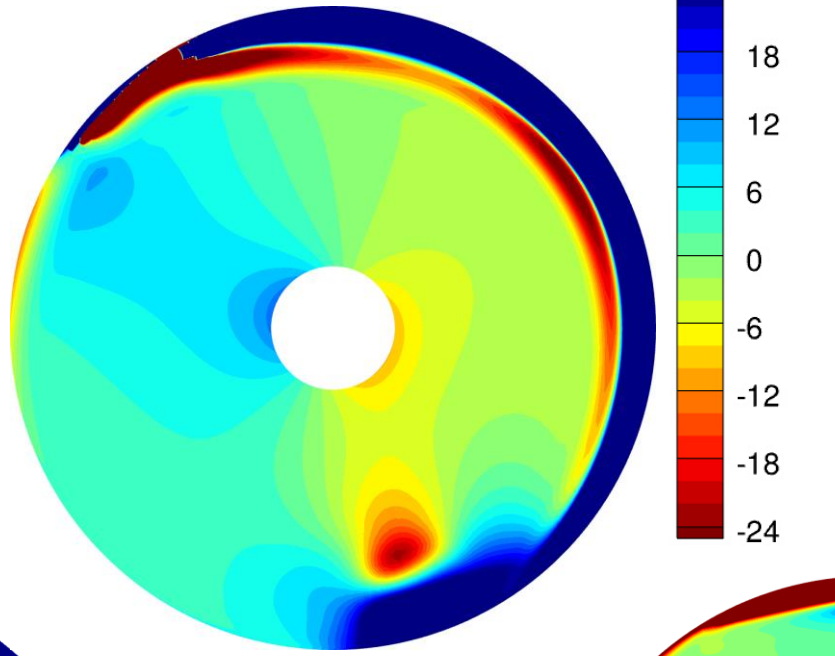
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LOW RPM SWIRL ANGLE

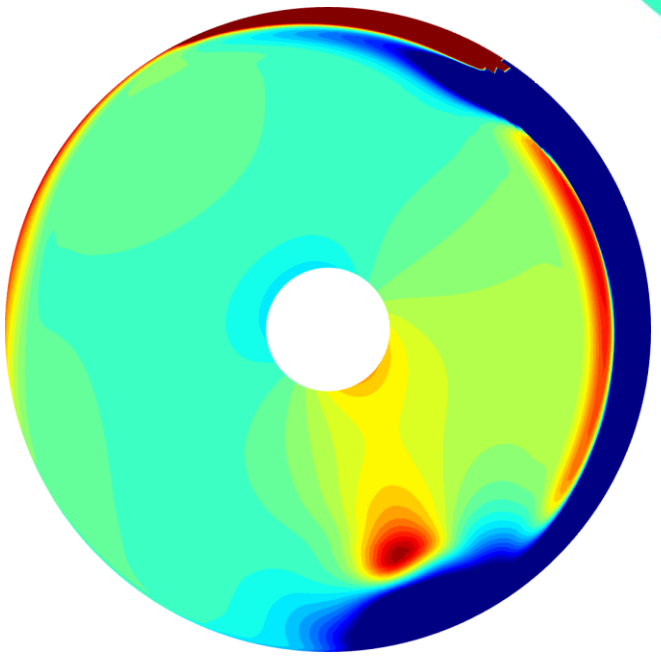


Piovesan et al., AIAA 2025-2111. January 2025.



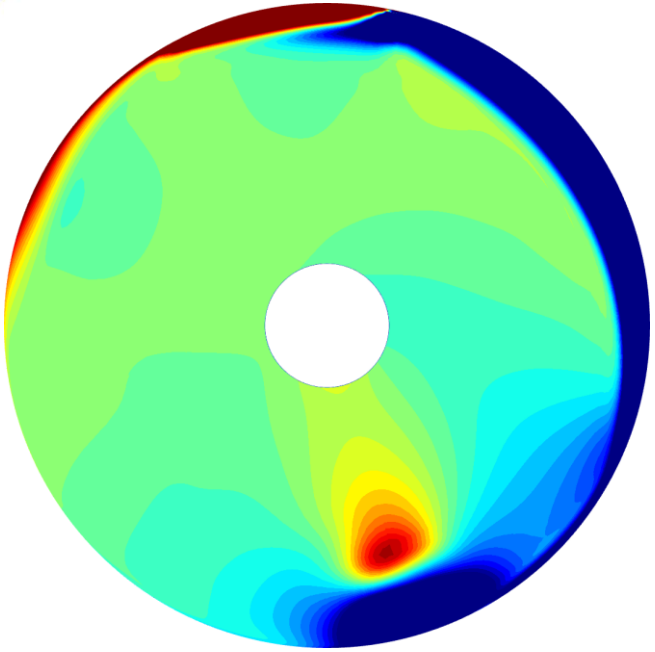


PDISK



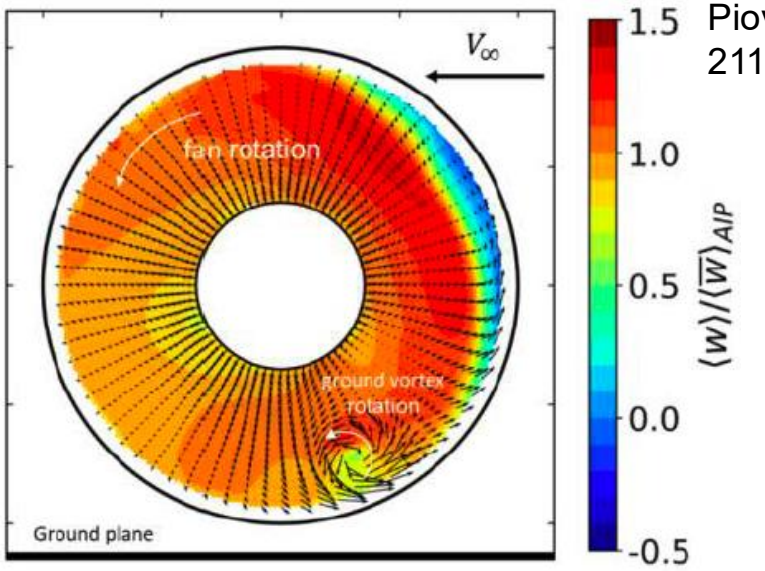
RDISK

URANS

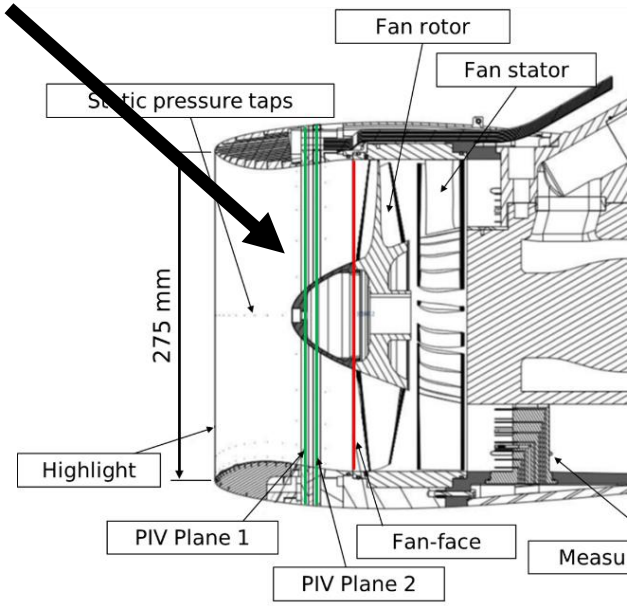


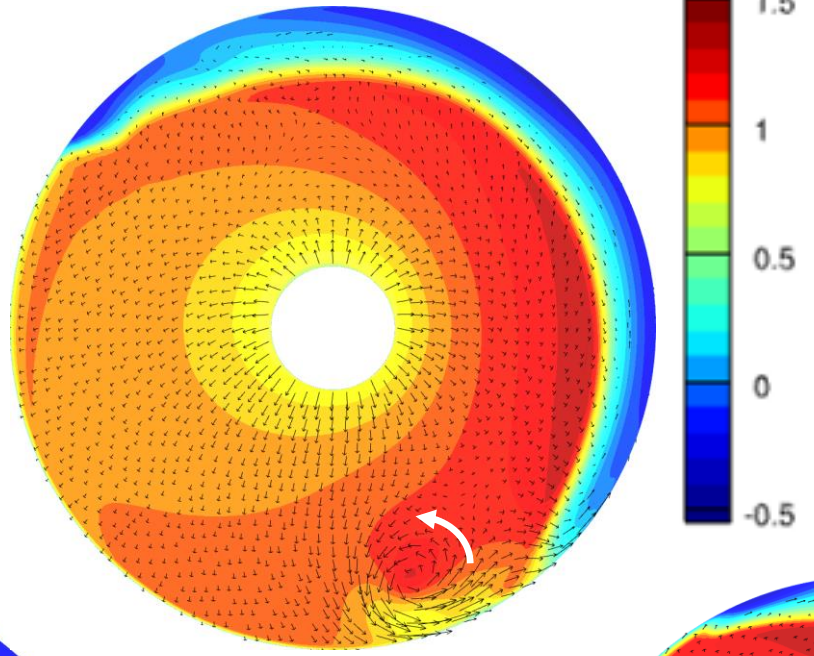
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LOW RPM AXIAL VELOCITY

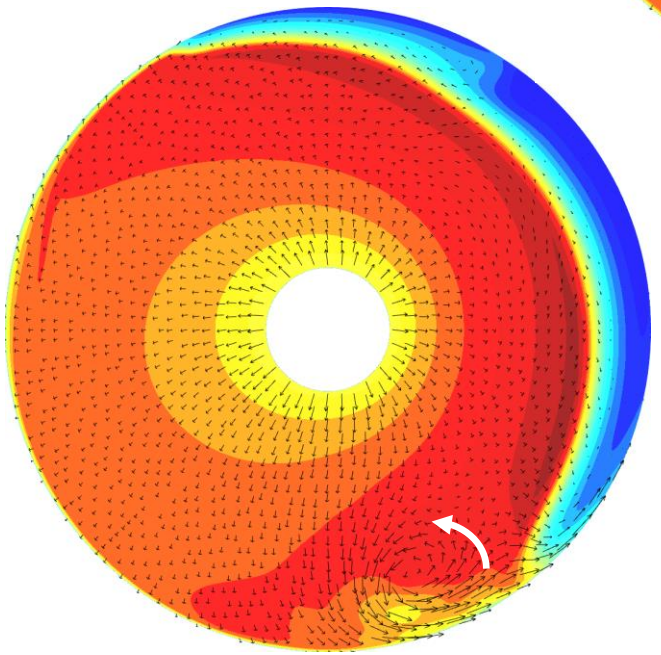


Piovesan et al., AIAA 2025-2111. January 2025.



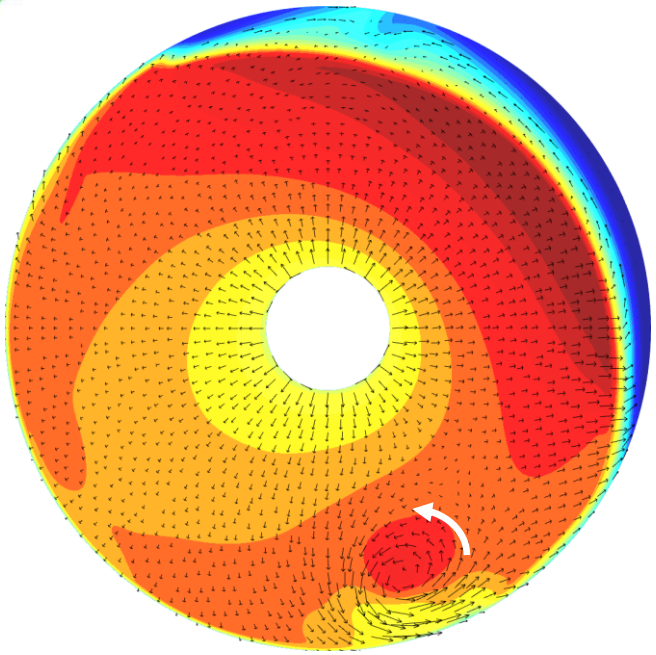


PDISK



RDISK

URANS



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CONCLUSIONS AND FUTURE WORK

- Outlined a plan of investigation for using rotor disks for modeling propulsors
- Showed preliminary results of coarse mesh computations
- Disk models thus far show good agreement for producing the ground vortex, but larger windward separation regions

Future work

- Mesh sensitivity/grid-refinement study
- Comparison with experimental data forthcoming
- PAW-08 geometry includes exit nozzle
- Blockage effects from high-solidity
- Also consider actuator line model

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ACKNOWLEDGEMENTS

Funding

NASA Subsonic Vehicle Technologies (SVT) Project

Computational Resources

NASA Advanced Supercomputing (NAS) facility

Collaborations, Discussions and Feedback

Anish Taylor (Boeing), Joe Derlaga and Chip Jackson (Overflow Team),
Aaron Burkhead (CAPE)

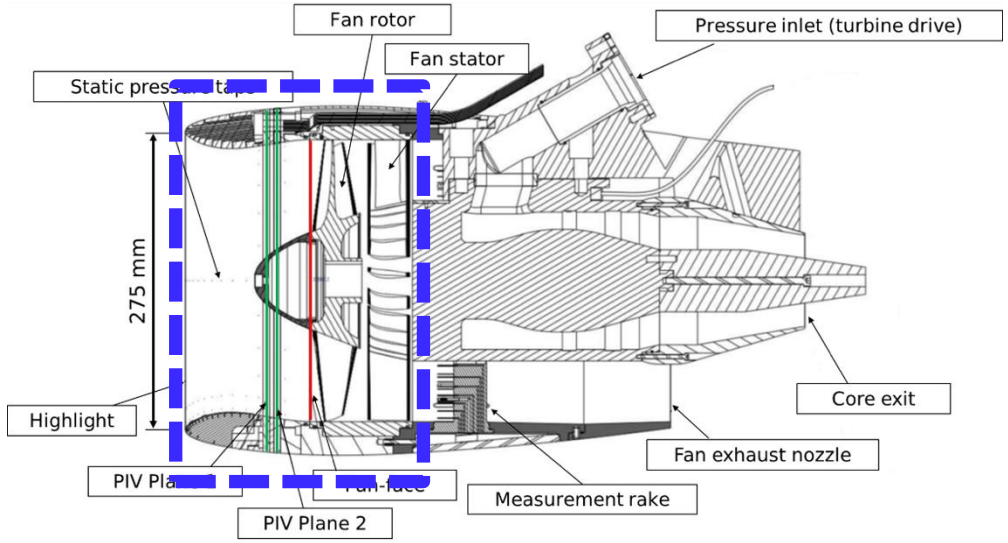
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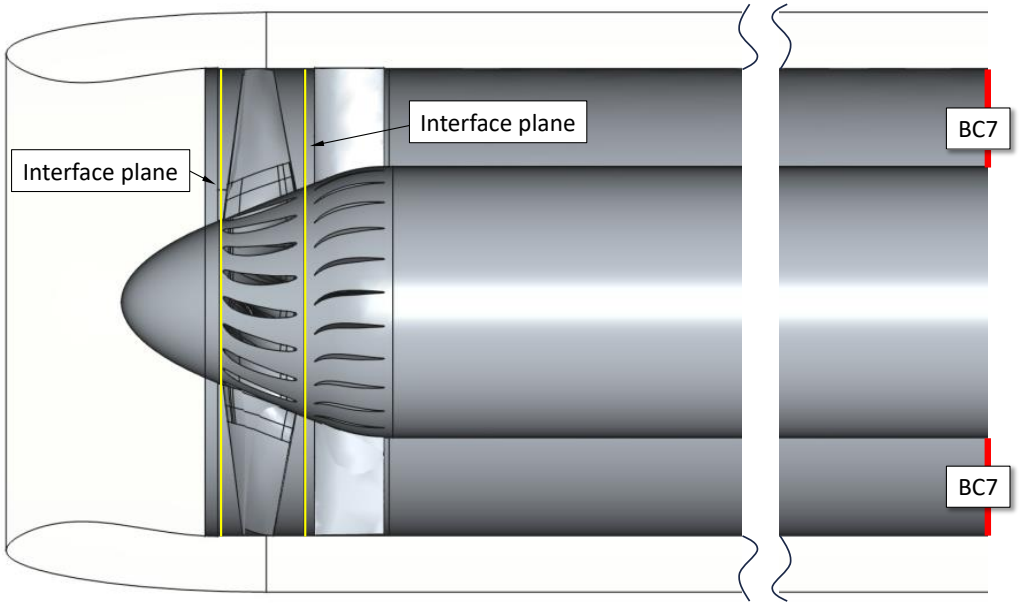
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BACKUP

SIMPLIFIED NIFTI GEOMETRY

- Flow through nacelle primarily driven by mass outflow BC
- Fan & stator primarily provide blockage effects and interaction with intake
- PAW 08 (not yet available) will include the fan exhaust nozzle
  - Fan RPM and exit area become relevant for mass flow





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BLADE-RESOLVED FAN (URANS)

- Fan & stator blade geometries fully included in computational domain, with unsteady solution
- Prescribed spinning motion applied to both fan blades and hub

